

Machine Learning Assignment 1: Polynomial Regression Analysis

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1 Introduction

This report details the implementation and evaluation of a polynomial regression model developed entirely from scratch using NumPy. The objective is to predict the continuous target variable y for two distinct scenarios: Steam Turbine Optimization (var1) and Subterranean Thermal Reservoir Mapping (var2). The report discusses the mathematical foundations aligned with the course notation, feature selection methodologies, and the determination of the optimal polynomial degree through hyperparameter sweeping using both standard hold-out and K -Fold cross-validation.

2 Methodology and Mathematical Formulation

Following the notation established in the lectures, let the n^{th} data point be denoted by the input vector $\mathbf{x}_n$ and the true target value as y_n (often denoted t_n in standard literature). The predictor or hypothesis function is parameterized by weights $\mathbf{w}$ and is defined as:

$$f(\mathbf{x}_n; \mathbf{w}) = \mathbf{w}^T \mathbf{x}_n \quad (1)$$

For polynomial regression of degree d with D original features, we expand the input vector $\mathbf{x}_n$ to include all combinations of polynomial terms (monomials) up to degree d .

The model is optimized by minimizing the averaged Squared Loss function $\mathcal{L}$ over all N data points:

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \sum_{n=1}^N \mathcal{L}_n(y_n, f(\mathbf{x}_n; \mathbf{w})) = \frac{1}{N} \sum_{n=1}^N (y_n - \mathbf{w}^T \mathbf{x}_n)^2 \quad (2)$$

Let $\mathbf{X}$ be the data matrix where the n^{th} row is $\mathbf{x}_n^T$, and let $\mathbf{y}$ be the column vector of targets. Setting the gradient of $\mathcal{L}(\mathbf{w})$ with respect to $\mathbf{w}$ to zero yields the Normal Equation:

$$\mathbf{X}^T \mathbf{X} \mathbf{w} = \mathbf{X}^T \mathbf{y} \quad (3)$$

Solving for $\mathbf{w}$, we obtain the closed-form optimal weights:

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (4)$$

To ensure numerical stability when $\mathbf{X}^T \mathbf{X}$ is ill-conditioned (which is common in high-degree polynomial expansions), the implementation calculates the least-squares solution using Singular Value Decomposition via NumPy.

3 Problem 1: Steam Turbine Optimization (var1)

3.1 Experimental Setup

The objective is to predict the Net Power Score (y) based on 6 operational parameters ($x_1, \dots, x_6$). We performed two exhaustive sweeps across all subsets of features and polynomial degrees (from 1 to 10): first using a standard 80/20 train-validation split, and subsequently using a rigorous 10-Fold Cross-Validation to assess model stability.

3.2 80/20 Split Analysis

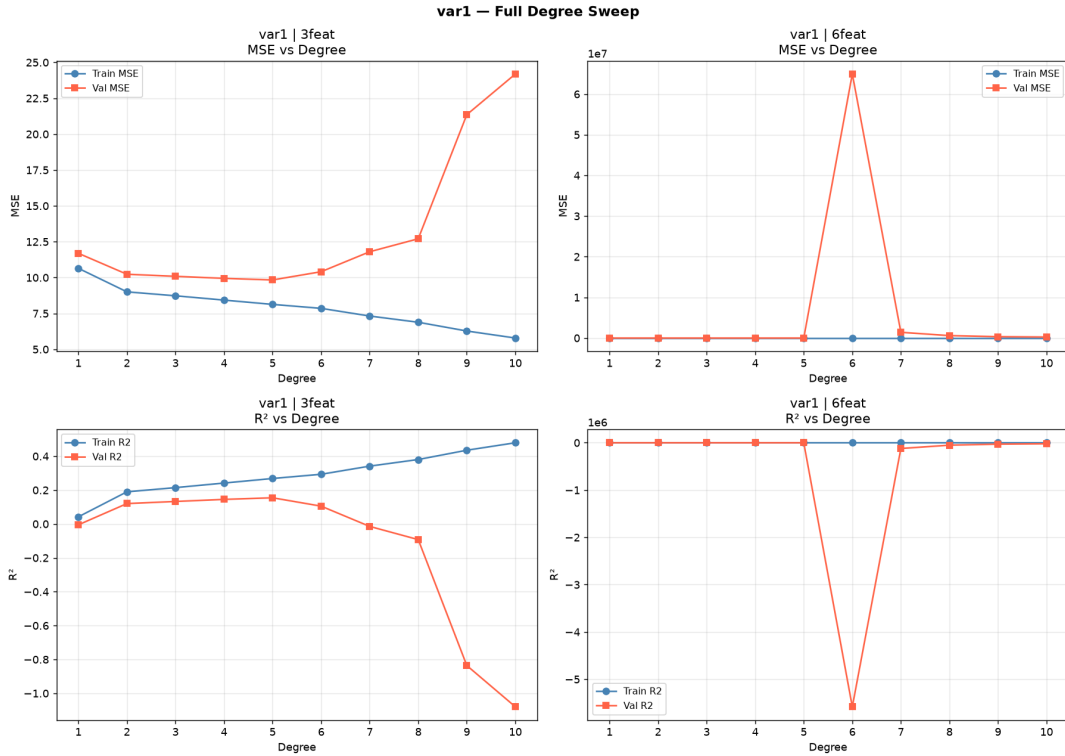


Figure 1: Degree Sweep for Problem 1 comparing different feature subsets (80/20 Split).

Degree	3 Features (x_1, x_2, x_3)		6 Features ($x_1 \dots x_6$)	
	Val MSE	Val R^2	Val MSE	Val R^2
2	10.23	0.120	3.46	0.703
3	10.09	0.133	1.12	0.904
4	9.95	0.145	0.835	0.928
5	9.83	0.155	1.21	0.896
6	10.41	0.105	6.49×10^7	Explodes

Table 1: Validation performance across polynomial degrees (80/20 split) for Problem 1.

As illustrated in Figure 1 and Table 1, omitting features (e.g., using only 3 features) results in severe underfitting. Incorporating all 6 features yields an optimal validation

MSE at **Degree 4** (MSE = 0.835). At Degree 6, the model suffers from extreme overfitting due to the curse of dimensionality.

3.3 10-Fold Cross Validation Analysis

To robustly verify this hyperparameter selection, a 10-Fold CV was applied to all 630 configurations (63 feature subsets $\times$ 10 degrees). The standard deviation of the validation MSE across the 10 folds was recorded as a metric of model stability.

Features Included	Degree	Avg Val MSE	Avg Val R^2	Val MSE Std Dev
$x_1, x_2, x_3, x_4, x_5, x_6$	3	1.108	0.901	0.235
$x_1, x_2, x_3, x_4, x_5, x_6$	4	0.833	0.924	0.144
$x_1, x_2, x_3, x_4, x_5, x_6$	5	1.014	0.911	0.347
x_1, x_2, x_3, x_5, x_6 (No x_4)	4	1.753	0.841	0.224

Table 2: Top configurations from 10-Fold Cross-Validation for Problem 1.

The 10-Fold CV (Table 2) definitively confirms that **Degree 4 with all 6 features** is superior. Not only does it hold the lowest average error (MSE = 0.833), but it is also the most stable configuration, exhibiting the lowest variance across folds (Std Dev = 0.144).

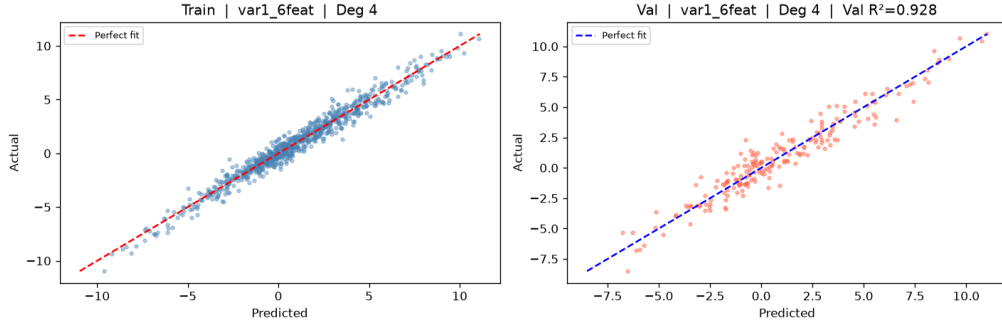


Figure 2: Residual Analysis for the optimal Degree 4 model (6 features).

4 Problem 2: Subterranean Thermal Reservoir Mapping (var2)

4.1 Experimental Setup

The objective is to predict the Thermal Anomaly Score (y) using 3 spatial coordinate offsets (x_1, x_2, x_3). As with Problem 1, we evaluated models using both an 80/20 split and exhaustive 10-Fold CV across all possible 70 feature/degree permutations.

4.2 80/20 Split Analysis

The initial 80/20 sweep (Figure 3 and Table 3) reveals that using a single feature completely fails to capture the target variance. With 3 features, the model steadily improves, minimizing validation error at **Degree 8**.

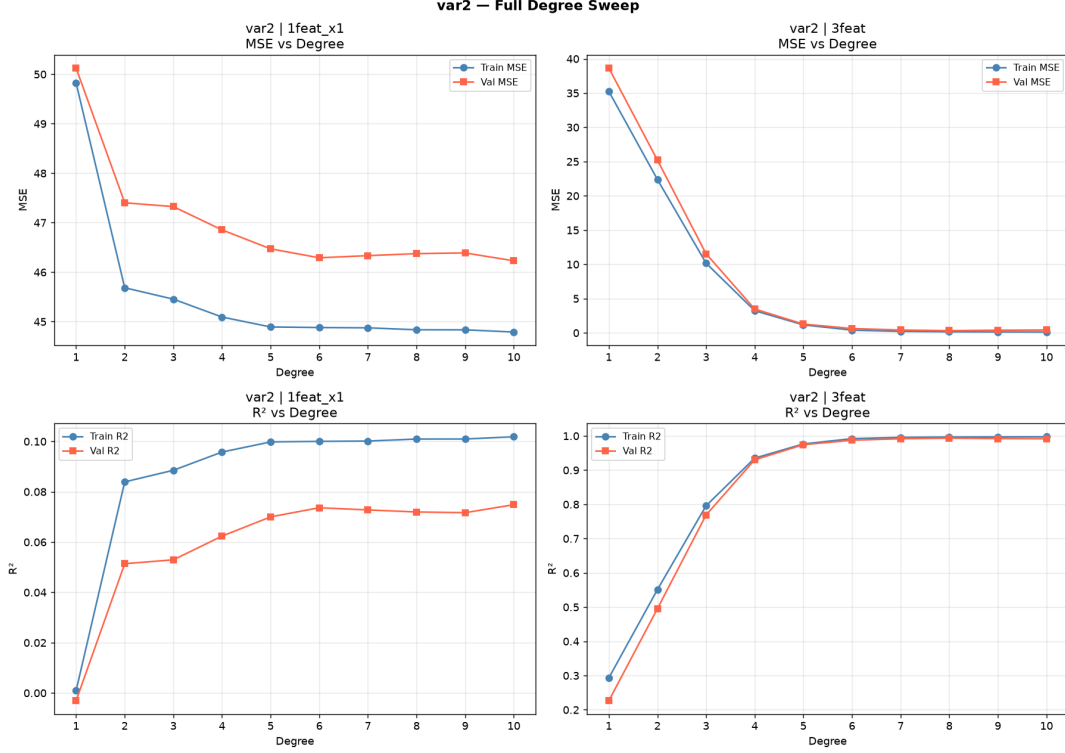


Figure 3: Degree Sweep for Problem 2 comparing different feature subsets.

Degree	1 Feature (x_1)		3 Features (x_1, x_2, x_3)	
	Val MSE	Val R^2	Val MSE	Val R^2
4	46.86	0.062	3.48	0.930
6	46.29	0.074	0.633	0.987
8	46.38	0.072	0.324	0.993
10	46.23	0.075	0.416	0.992

Table 3: Validation performance across polynomial degrees (80/20 split) for Problem 2.

4.3 10-Fold Cross Validation Analysis

The 10-Fold CV results in Table 4 corroborate the 80/20 findings. At **Degree 8**, the validation MSE achieves its true minimum (0.273) alongside an exceptional stability metric (Std Dev = 0.027). Pushing the polynomial to Degree 10 drastically increases the standard deviation (0.196), indicating the onset of model variance.

The residual plot in Figure 4 confirms that the validation predictions tightly hug the perfect-fit line. Consequently, the **Degree 8 model with all 3 features** was definitively chosen as the final model for Problem 2.

5 Conclusion

Through systematic feature selection and hyperparameter sweeping via both 80/20 splits and rigorous 10-Fold Cross-Validation, optimal polynomial regression models were mathematically identified. The analysis demonstrated that leveraging the full feature set up

Features Included	Degree	Avg Val MSE	Avg Val R^2	Val MSE Std Dev
x_1, x_2, x_3	6	0.573	0.988	0.069
x_1, x_2, x_3	7	0.336	0.993	0.045
$\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$	8	0.273	0.994	0.027
x_1, x_2, x_3	9	0.297	0.994	0.028
x_1, x_2, x_3	10	0.401	0.992	0.196

Table 4: Top configurations from 10-Fold Cross-Validation for Problem 2.

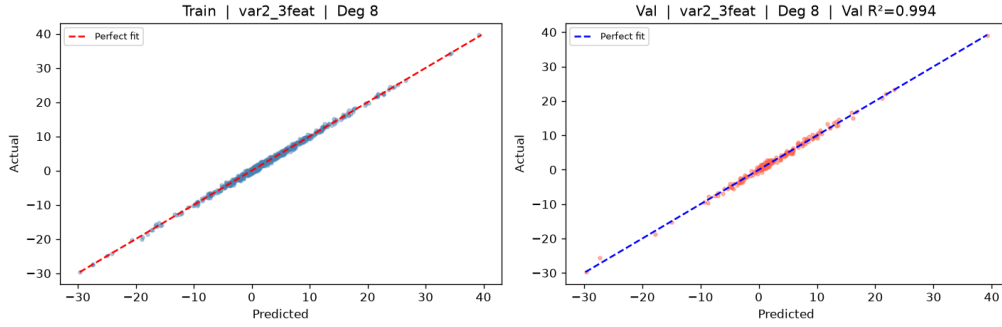


Figure 4: Residual Analysis for the optimal Degree 8 model (3 features).

to an optimal complexity boundary accurately minimizes the squared loss $\mathcal{L}(\mathbf{w})$ while maintaining model stability across holdout folds.